Module-3

IoT Processing Topologies and Types:

- Data Format,
- Importance of Processing in IoT,
- Processing Topologies,
- IoT Device Design and Selection Considerations,
- Processing Offloading.

Textbook 1: Chapter 6 – 6.1 to 6.5

3.1 Data Format

- The Internet is a vast space where huge quantities and varieties of data are generated regularly and flow freely.
- The massive volume of data generated by this huge number of users is further enhanced by the multiple devices utilized by most users.
- In addition to these data-generating sources, non-human data generation sources such as sensor nodes and automated monitoring systems further add to the data load on the Internet.
- This huge data volume is composed of a variety of data such as e-mails, text documents (Word docs, PDFs, and others), social media posts, videos, audio files, and images, as shown in Figure 6.1.
- Data can be broadly grouped into two types based on how they can be accessed and stored:

1) Structured data and
2) Unstructured data.

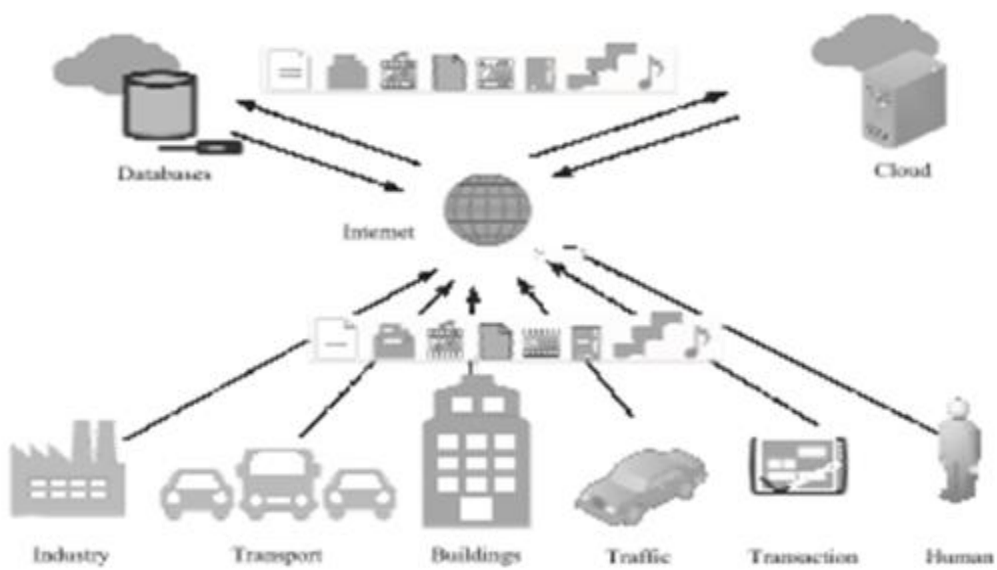

Figure 6.1 The various data generating and storage sources connected to the Internet and the plethora of data types contained within it

Structured data

- These are text data that have a pre-defined structure.
- Associated with relational database management systems (RDBMS).
- These are primarily created by using length-limited data fields such as phone numbers, social security numbers, and other such information.
- Even if the data is human or machine generated, these data are easily searchable by querying algorithms as well as human generated queries.
- Eg: flight or train reservation systems, banking systems.
- Established languages such as Structured Query Language (SQL) are used for accessing these data in RDBMS.

Unstructured data

- All the data on the Internet, which is not structured, is categorized as unstructured.
- These data types have no pre-defined structure and can vary according to applications and data-generating sources.
- Some of the common examples of human-generated unstructured data include text, e-mails, videos, images, phone recordings, chats, and others.
- Some common examples of machine-generated unstructured data include sensor data from traffic, buildings, industries, satellite imagery, surveillance videos, and others.
- This data type does not have fixed formats associated with it, which makes it very difficult for querying algorithms to perform a look-up.
- Querying languages such as NoSQL are generally used for this data type.

3.2 Importance of Processing in IoT

- The vast amount and types of data flowing through the Internet.
- The need for intelligent and resourceful processing techniques.
- More crucial with the rapid advancements in IoT,
- To process and what to process?
- The data to be processed into three types based on the urgency of processing:
    1) Very time critical,
    2) Time critical,
    3) Normal.

Very time critical,

- Data from sources such as flight control systems, healthcare, and other such sources, which need immediate decision support, are deemed as very critical.
- These data have a very low threshold of processing latency, typically in the range of a few milliseconds.

Time critical

- Data from sources that can tolerate normal processing latency are considered as time critical data.
- These data, generally associated with sources such as vehicles, traffic, machine systems, smart home systems, surveillance systems, and others, which can tolerate a latency of a few seconds fall in this category.

Normal

- Can tolerate a processing latency of a few minutes to a few hours and are typically associated with less data-sensitive domains such as agriculture, environmental monitoring, and others.

3.3 Processing Topologies

- The architecture of the deployment is required for intelligent selection of processing requirement of an IoT application.
- A properly designed IoT architecture would result in
  - massive savings in network bandwidth
  - conserve significant amounts of overall energy in the architecture
  - allowable processing latencies for the solutions
- The various processing solutions into two large topologies:

1) On-site

2) Off-site.

- The off-site processing topology can be further divided into the following:

1) Remote processing

2) Collaborative processing.

On-site processing

- The data is processed at the source itself.
- Crucial in applications that have a very low tolerance for latencies.
- The latencies may result from the processing hardware or the network (during transmission of the data for processing away from the processor).
- Applications such as those associated with healthcare and flight control systems(real time systems) have a breakneck data generation rate.
- These additionally show rapid temporal changes that can be missed (leading to catastrophic damages) unless the processing infrastructure is fast and robust enough to handle such data.
- Figure 6.2 shows the on-site processing topology, where an event (here, fire) is detected utilizing a temperature sensor connected to a sensor node.

- The sensor node processes the information from the sensed event and generates an alert. The node additionally has the option of forwarding the data to a remote infrastructure for further analysis and storage.

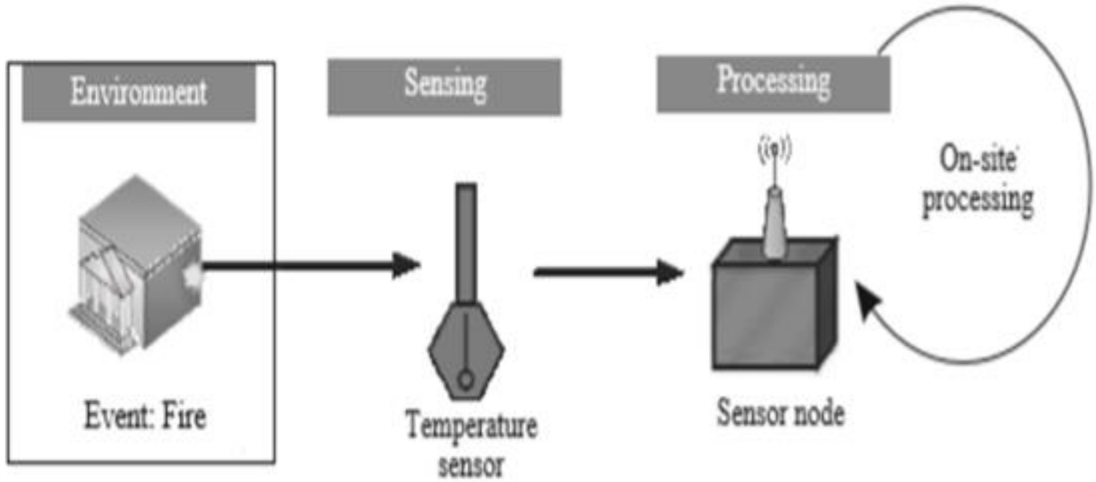

Figure 6.2 Event detection using an on-site processing topology

Off-site processing

- Opposed to the on-site processing paradigms, allows for latencies (due to processing or network latencies);
- It is significantly cheaper than on-site processing topologies.
- This difference in cost is mainly due to the low demands and requirements of processing at the source itself.
- Sensor nodes are not required to process data on an urgent basis.
- Dedicated and expensive on-site processing infrastructure is not sustainable for large-scale deployments typical of IoT deployments.
- In the off-site processing topology, the sensor node is responsible for the collection and framing of data to be transmitted to another location for processing.

- Unlike the on-site processing topology, the off-site topology has a few dedicated high-processing enabled devices, which can be borrowed by multiple
- In the off-site topology, the data from these sensor nodes (data generating sources) is transmitted either to a remote location (which can either be a server or a cloud) or to multiple processing nodes.

Remote processing

- Most common processing topologies in present-day IoT solutions.
- It encompasses sensing of data by various sensor nodes; the data is then forwarded to a remote server or a cloud-based infrastructure for further processing and analytics.
- The processing of data from hundreds and thousands of sensor nodes can be simultaneously off loaded to a single, powerful computing platform;
- This results in massive cost and energy savings by enabling the reuse and reallocation of the same processing resource while also enabling the deployment of smaller and simpler processing nodes at the site of deployment.
- Figure 6.3 shows the outline of one such paradigm, where the sensing of an event is performed locally, and the decision making is out sourced to are mote processor (here, cloud).

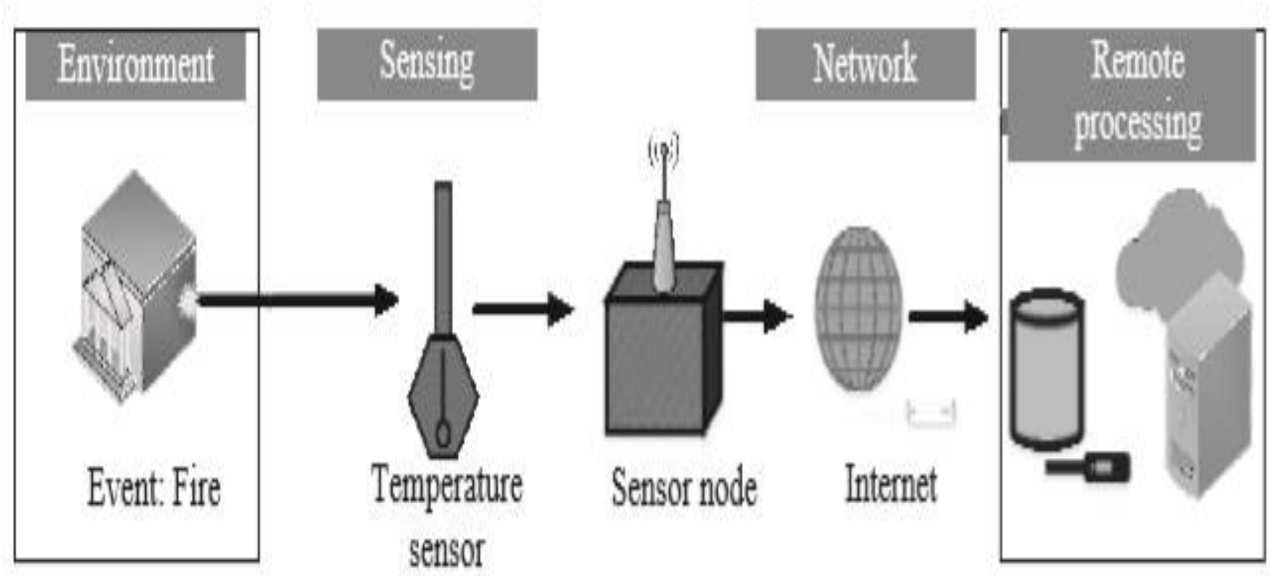

Figure 6.3 Event detection using an off-site remote processing topology

Collaborative processing

- This processing topology typically finds use in scenarios with limited or no network connectivity, especially systems lacking a backbone network.
- Quite economical for large-scale deployments spread over vast areas, where providing networked access to a remote infrastructure is not viable.
- The simplest solution is to club together the processing power of nearby processing nodes and collaboratively process the data in the vicinity of the data source itself.
- This approach also reduces latencies due to the transfer of data over the network.
- Conserves bandwidth of the network, especially ones connecting to the Internet.
- Figure 6.4 shows the collaborative processing topology for collaboratively processing data locally.
- This topology can be quite beneficial for applications such as agriculture.
- Mesh networks for easy implementation of this topology.

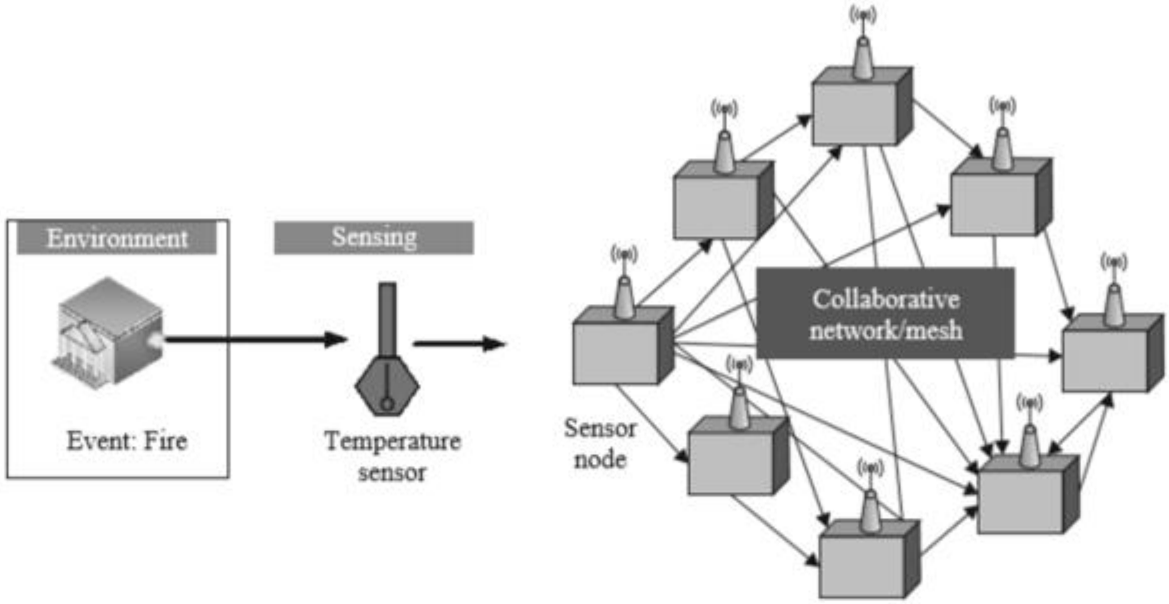

Figure 6.4 Event detection using a collaborative processing topology

3.4 IoT Device Design and Selection Considerations

- The main consideration of minutely defining an IoT solution is the selection of the processor for developing the sensing solution (i.e., the sensor node).
- This selection isgoverned by many parameters that affect the usability, design, and affordability of the designed IoT sensing and processing solution.
- The main factor governing the IoT device design and selection for various applications is the processor.
- The other important considerations are as follows.

Size:

- This is one of the crucial factors for deciding the form factor and theenergy consumption of a sensor node.
- Larger the formfactor, larger is the energy consumption of the hardware.
- Additionally, large formfactors are not suitable for a significant bulk of IoT applications, which rely onminimal form factor solutions (e.g., wearables).

Energy:

- The energy requirements of a processor are the most importantdeciding factor in designing IoT-based sensing solutions.
- Higher the energyrequirements, higher is the energy source (battery) replacement frequency.

Cost:

- The cost of a processor, besides the cost of sensors, is the driving forcein deciding the density of deployment of sensor nodes for IoT-based solutions.
- Cheaper cost of the hardware enables a much higher density of hardware deployment by users of an IoT solution.
- For example, cheaper gas and firedetection solutions would enable users to include much more sensing hardware for a lesser cost.

Memory:

- The memory requirements (both volatile and non-volatile memory) o fIoT devices determines the capabilities the device can be armed with.
- Featuressuch as local data processing, data storage, data filtering, data formatting, and a host of other features rely heavily on the memory capabilities of devices.
- However, devices with higher memory tend to be costlier for obvious reasons.

Processing power:

- Processing power is vital(comparable to memory) in deciding what type of sensors can be accommodated with the IoT device/node, and what processing features can integrate on-site with the IoT device.
- The processing power also decides the type of applications.
- Typical applications that handle video and image data require IoT devices with higher processing power as compared to applications requiring simple sensing of the environment.

I/O rating:

- The input–output (I/O) rating of IoT device, primarily the processor,is the deciding factor in determining the circuit complexity, energy usage, andrequirements for support of various sensing solutions and sensor types.
- Processors have a meager I/O voltage rating of 3.3 V, as compared to 5 V for the somewhat older processors.
- This translates to requiring additional voltage andlogic conversion circuitry to interface legacy technologies and sensors with thenewer processors.
- Despite low power consumption due to reduced I/O voltagelevels, this additional voltage and circuitry not only affects the complexity of thecircuits but also affects the costs.

Add-ons:

- The support of various add-ons a processor an IoT device provides - analog to digital conversion (ADC) units, in-built clockcircuits, connections to USB and ethernet, inbuilt wireless access capabilities, andothers helps in defining the robustness and usability of a processor or IoT device in various application scenarios.
- The provision for these add-onsalso decides how fast a solution can be developed, especially the hardware part of the whole IoT application.
- As interfacing and integration of systems at thecircuit level can be daunting to the uninitiated, the prior presence of these optionswith the processor makes the processor or device highly lucrative to the users/developers.

3.5 Processing Offloading

- The processing offloading paradigm is important for the development of densely deployable, energy-conserving, miniaturized, and cheap IoT-based solutions for sensing tasks.
- Figure 6.5 shows the typical outline of an IoT deployment with the various layers ofprocessing that are encountered spanning vastly different application domains—from as near as sensing the environment to as far as cloud-based infrastructure.
- The sensors enabling the sesensing types are integrated with a processor using wired or wireless connections.
- In the event that certain applications require immediate processing of the sensed data, an on-site processing topology is followed.
- The majority of IoT applications, the bulk of the processing is carriedout remotely in order to keep the on-site devices simple, small, and economical.
- For off-site processing, data from the sensing layer can be forwarded tothe fog or cloud or can be contained within the edge layer.
- The devices within the local network, till the fog communicate using short-range wireless connections.
- In case the data needs to be sent further up the chain to the cloud, long-range wireless connection enabling access to a backbone network is essential.
- Fog nodes, which are at the level of gateways, may or may not be accessed the IoT devices through the Internet.
- The approach of forwarding data to a cloud or a remote server, as shown in the topology in Figure 6.3.
- Requires the devices to be connected to the Internetthrough long-range wireless/wired networks, which eventually connect to a backbonenetwork.
- This approach is generally costly concerning network bandwidth, latency, aswell as the complexity of the devices and the network infrastructure involved.

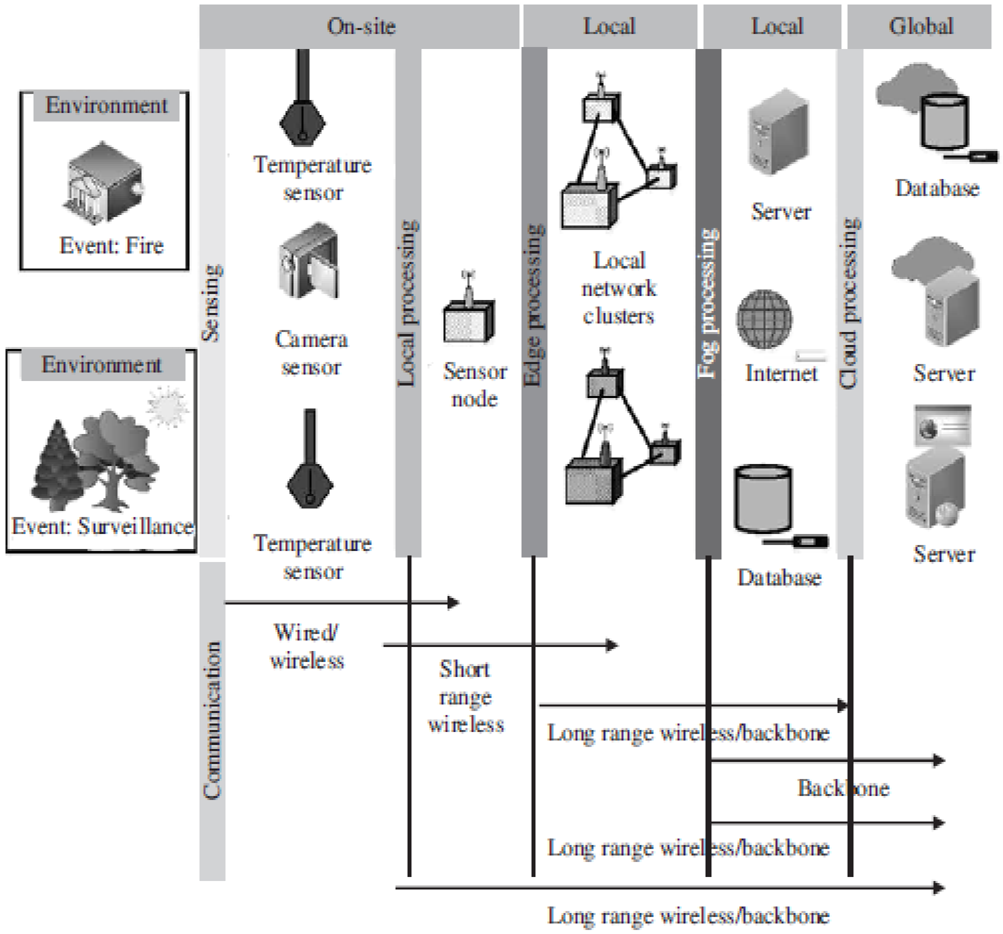

Figure 6.5 The various data generating and storage sources connected to the Internet and the plethora of data types contained within it

- Data offloading is divided into three parts:

  1) Offload location(which outlines where all the processing can be offloaded in the IoT architecture),

  2) Offload decision making (how to choose where to offload the processing to and by How much), and finally

  3) Offloading considerations (deciding when to offload).

3.5.1 Offload location

- The choice of offload location decides the applicability, cost, and sustainability of the IoT application and deployment.
- The offload location into four types:

1. Edge
2. Fog
3. Remote Server
4. Cloud

Edge:

- Offloading processing to the edge implies that the data processing is facilitated to a location at or near the source of data generation itself.
- Offloading to the edge is done to achieve aggregation, manipulation, bandwidth reduction and other data operations directly on an IoT device.

Fog:

- Fog computing is a decentralized computing infrastructure that is utilized to conserve network bandwidth, reduce latencies, restrict the amount of data unnecessarily flowing through the Internet, and enable rapid mobility support for IoT devices.
- The data, computing, storage and applications are shifted to a place between the data source and the cloud resulting in significantly reduced latencies and network bandwidth usage.

Remote Server:

- A simple remote server with good processing power maybe used with IoT-based applications to offload the processing from resource-constrained IoT devices.
- Rapid scalability may be an issue with remote servers and they may be costlier and hard to maintain in comparison to solutions such as the cloud.

Cloud:

- Cloud computing is a configurable computer system which can get access to configurable resources, platforms, and high-level services through a shared pool hosted remotely.

- A cloud is provisioned for processing offloading so that processing resources can be rapidly provisioned with minimal effort over the Internet, which can be accessed globally.
- Cloud enables massive scalability of solutions as they can enable resource enhancement allocated to a user or solution in an on-demand manner, without the user having to go through the pains of acquiring and configuring new and costly hardware.

3.5.2 Offload decision making

- The choice of where to offload and how much to offload is one of the major deciding factors in the deployment of an offsite-processing topology-based IoT deployment architecture.
- The decision-making is generally addressed considering data generation rate, network bandwidth, the criticality of applications, processing resource available at the offload site and other factors.
- Some of these approaches are as follows.

Naive Approach:

- This approach is typically a hard approach, without too much decision making.
- It can be considered as a rule-based approach in which the data from IoT devices are off loaded to the nearest location based on the achievement of certain offload criteria.
- Easy to implement, this approach is never recommended, especially for dense deployments, or deployments where the data generation rate is high or the data being off loaded in complex to handle (multimedia or hybrid data types).
- Generally, statistical measures are consulted for generating the rules for offload decision making.

Bargaining based approach:

- This approach, although a bit processing-intensive during the decision making stages, enables the alleviation of network traffic congestion, enhances service QoS (quality of service) parameters such as bandwidth, latencies, and others.
- Maximize multiple parameters for the whole IoT implementation to provide the most optimal solution or QoS.
- Bargaining based solutions try to maximize the QoS by trying to reach a point where the qualities of certain parameters are reduced, while the others are enhanced.
- This measure is undertaken so that the achieved QoS is collaboratively better for the full implementation rather than a select few devices enjoying very high QoS.
- Game theory is a common example of the bargaining based approach.
- This approach does not need to depend on historical data for decision making purposes.

Learning based approach:

- The learning based approaches generally rely on past behavior and trends of data flow through the IoT architecture.
- The optimization of QoS parameters is pursued by learning from historical trends and trying to optimize previous solutions further and enhance the collective behavior of the IoT implementation.
- The memory requirements and processing requirements are high during the decision making stages.
- The most common example of a learning based approach is machine learning.

6.5.3 Offloading considerations

- There are a few offloading parameters which need to be considered while deciding upon the offloading type to choose.
- These considerations typically arise from the nature of the IoT application and the hardware being used to interact with the application.
- Some of these parameters are as follows.

- **Bandwidth:** The maximum amount of data that can be simultaneously transmitted over the network between two points is the bandwidth of that network. The bandwidth of a wired or wireless network is also considered to be its data-carrying capacity and often used to describe the data rate of that network.
- **Latency:** It is the time delay incurred between the start and completion of an operation. In the present context, latency can be due to the network (network latency) or the processor (processing latency). In either case, latency arises due to the physical limitations of the infrastructure, which is associated with an operation. The operation can be data transfer over a network or processing of a data at a processor.
- **Criticality:** It defines the importance of a task being pursued by an IoT application. The more critical a task is, the lesser latency is expected from the IoT solution. For example, detection of fires using an IoT solution has higher criticality than detection of agricultural field parameters. The former requires a response time in the tune of milliseconds, whereas the latter can be addressed within hours or even days.
- **Resources:** It signifies the actual capabilities of an offload location. These capabilities may be the processing power, the suite of analytical algorithms, and others. For example, it is futile and wasteful to allocate processing resources reserved for real-time multimedia processing (which are highly energy-intensive and can process and analyze huge volumes of data in a short duration) to scalar data (which can be addressed using nominal resources without wasting much energy).
- **Data volume**: The amount of data generated by a source or sources that can be simultaneously handled by the offload location is referred to as its data volume handling capacity. Typically, for large and dense IoT deployments, the offload location should be robust enough to address the processing issues related to massive data volumes.